



**APD: AVALANCH PHOTO DIODE
TRAINER KIT**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the leading subjects of this century & expanding its horizon in coming century too.

Whether fiber optic system are for transmitting voice, video & audio or training regardless of application, it is our vision to provide you with the product you need, ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual, you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook for any further help.

After unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In **Fiber Optic Communication** modules, signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180 Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solution's sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solution's judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions. bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

APD : 2.5 Gbps InGaAs APD module

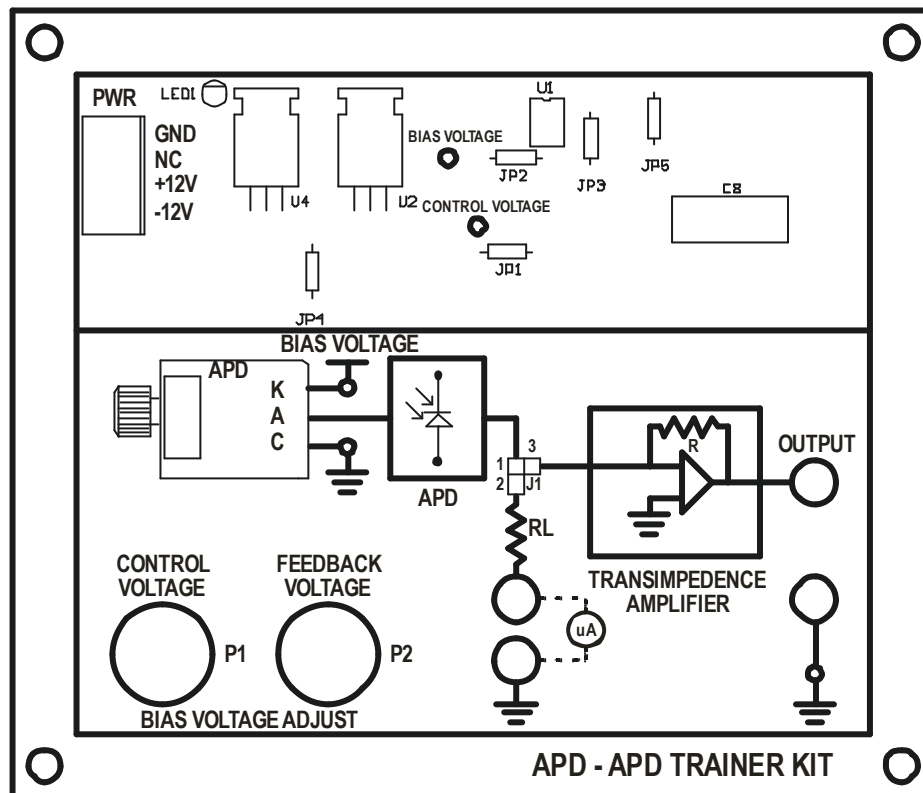
Specifications (T=25°C, -5V)

Parameter	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Responsivity	R	850nm, Vr=40V	0.15	-	-	A/W
		1310nm, Vr=10V	0.75	-	-	
		1550nm, Vr=10V	0.85	-	-	
Spectral Range		-	850	-	1650	nm
Breakdown Voltage	V_{bd}	$I_d=10\mu A$	40	50	65	V
Temperature Coefficient of Breakdown Voltage			-	0.059	-	V/°C
Dark Current	I_d	$0.9V_{bd}, 25^\circ C$	-	50	-	nA
	I_d	$0.9V_{bd}, 85^\circ C$	-	500	-	nA
Gain	M	$0.9V_{bd}$	7	10		
Terminal Capacitance	C	$0.9V_{bd}, f=1MHz$	-	0.35	0.5	pF
Bandwidth (3dB)	BW	M=10	3	3.5	4	GHz
Series Resistance	R_s		-	-	60	Ω
Excess Noise Factor	F	M=10	-	3	4	
Return Loss						
-1(Receptacle)			14	-	-	dB
-3(Receptacle or Pigtail)			30			dB
-4(Receptacle or Pigtail)			40	-	-	dB
-5(Pigtail)			50	-	-	dB

ABSOLUTE MAXIMUM RATINGS:

	Symbol	Min.	Max.	Unit
Operating Temperature	T_o	-40	85	°C
Storage Temperature	T_{stg}	-40	100	°C
Forward Current	I_F	-	10	mA
Reverse Voltage	V_R	-	V_{bd}	V
Reverse Current	I_R	-	3	mA
Optical Input	P	-	0	dBm
Lead Soldering Temperature (10 sec)	T_L	-	260	°C

Functional Block Diagram



INTRODUCTION

Fiber optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprints have saturated the airways & print media with advertisements heralding this & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

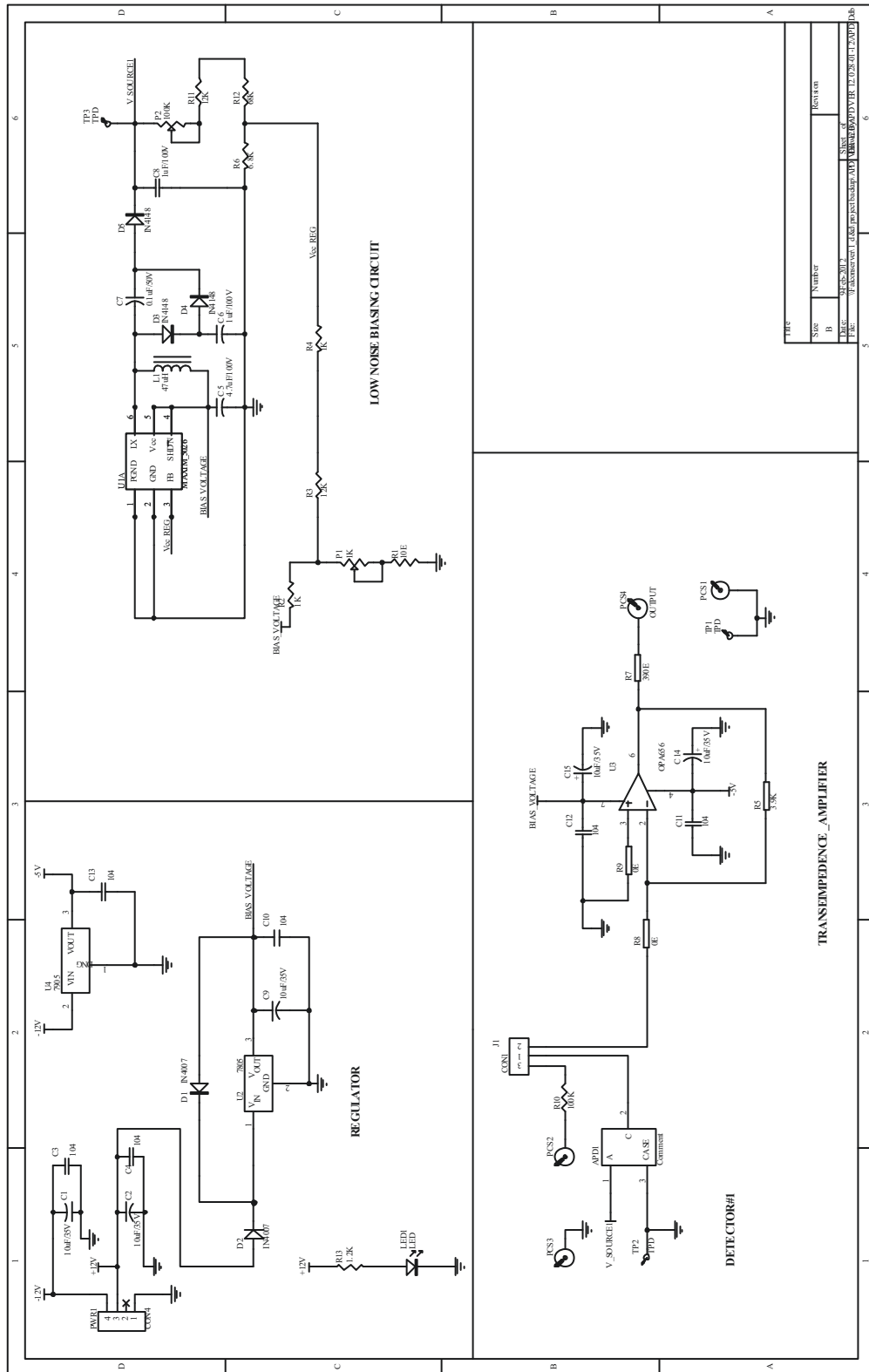
And their enthusiasm is valid. Today medical fiber optics systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes workspaces & recreational facilities. Your television reception will take on starling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optics will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few experts exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons existing with the prospect of making new discoveries finding new application for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & -yes- even fun.

APD circuit diagram



ABOUT APD

Avalanche Photodiodes (APDs) are high sensitivity, high speed semi-conductor “light” sensors. Compared to regular PIN construction photodiodes, APDs, have an internal region where electron multiplication occurs, by application of an external reverse voltage, and the resultant “gain” in the output signal means that low light levels can be measured at high speed. Incident photons create electron – hole pairs in the depletion layer of a silicon photodiode structure and these move towards the respective PN junctions at a speed of up to 10^5 metres per second, depending on the electric field strength. If the external bias increases this localised electric field to above about 10^5 V / cm then the carriers in the semi-conductor collide with atoms in the crystal lattice, and the resultant ionization creates more electron – hole pairs, some of which then go on to cause further ionization giving a resultant gain in the number of electron – holes generated for a single incident photon (See schematic below).

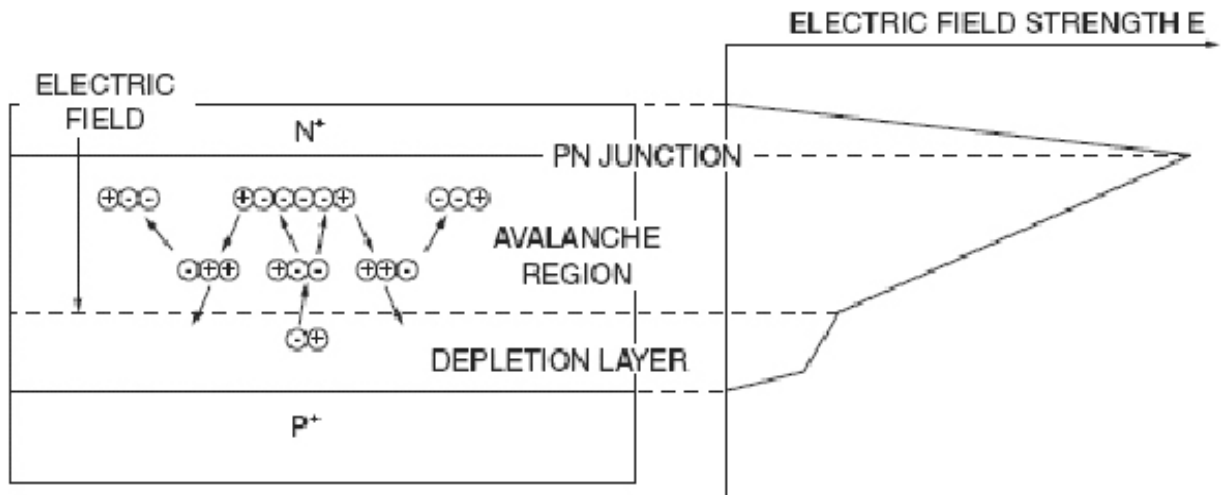


Figure 1-1 Schematic diagram of avalanche process

1-1 Principle of Avalanche Multiplication

When light enters a photodiode, electron-hole pairs are generated if the light energy is higher than the band gap energy. Light energy E (eV) and wavelength λ (nm) have a particular relation as shown in Equation 1-1 below.

$$\lambda = 1240/E \text{ (nm)} \dots\dots\dots (1.1)$$

The band gap of an InGa is less than 1.12 eV at room temperatures, so that it is sensitive to light wavelengths shorter than 1100 nm. This sensitivity is commonly expressed by terms called photosensitivity S (A/W) and quantum efficiency QE (%). The photosensitivity is the photocurrent divided by the incident radiant power, expressed in A/W. The quantum efficiency is the ratio of electron- hole

pairs generated versus the number of incident photons. These two terms have the following relation.

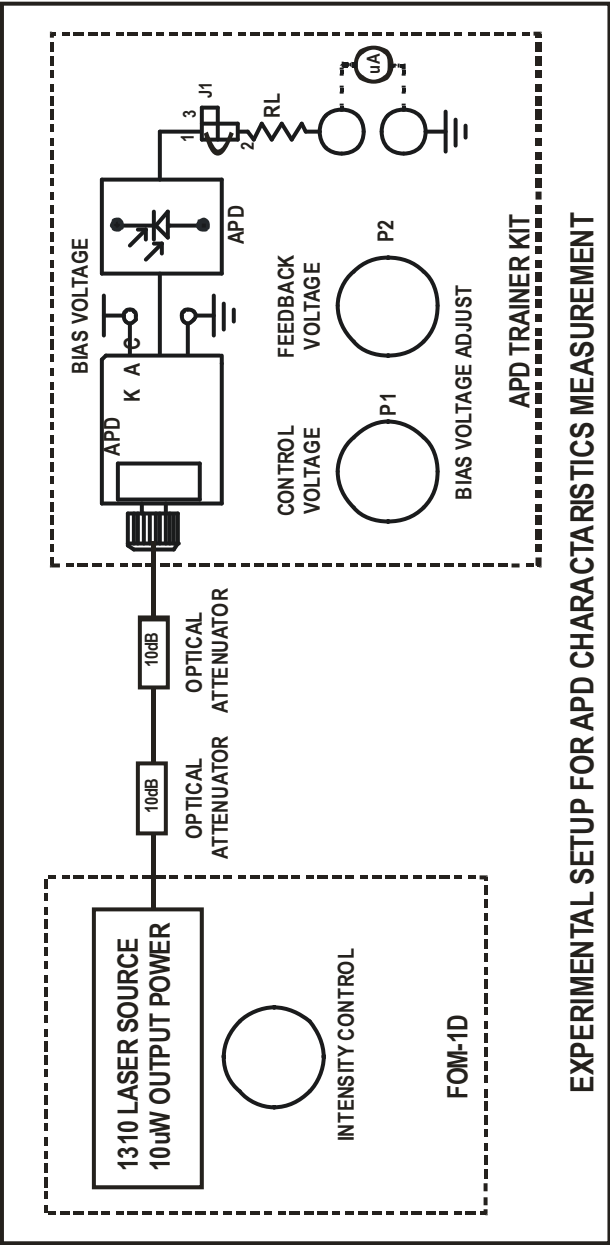
$$QE = S \times 1240 / \lambda \times 100 \% \dots\dots\dots (1.2)$$

When electron-hole pairs are generated in the depletion layer of a photodiode with a reverse voltage applied to the PN junction, the electrons drift towards the N+ side while the holes drift towards the P+ side due to the electric field developed across the PN junction. The drift speed of these electron-hole pairs or carriers depends on the electric field strength. However, when the electric field is increased to a certain level, the carriers are more likely to collide with the crystal lattice so that their drift speed becomes saturated at an average speed. This phenomenon begins to occur when the electric field is in the vicinity of 10^4 V/cm, and the saturated drift speed at this point is about 10^7 cm/s. If the reverse voltage is increased even further, some of the carriers which escaped collision with the crystal lattice will have a great deal of energy. When these carriers collide with the crystal lattice, ionization in which electron-hole pairs are newly generated takes place. These electron-hole pairs then create additional electron-hole pairs in a process just like a chain reaction. This is a phenomenon referred to as avalanche multiplication of photocurrent. This phenomenon begins

to take place when the electric field strength reaches 2×10^5 V/cm. Figure 1-1 is the schematic diagram of an avalanche process showing how the generated carriers are multiplied inside the APD. The number of electron-hole pairs generated during the time that the carriers travel a given distance is referred to as the ionization rate. Usually, the ionization rate of electrons is defined as α and that of holes as β . These ionization rates are important factors in determining the avalanche multiplication mechanism. The ratio k of β to α is called the ionization ratio and is used as a parameter to indicate device noise. (See Equation 1-3.)

$$K = \beta / \alpha \dots\dots\dots (1.3)$$

EXPERIMENT NO.1



EXPERIMENT NO.1

NAME

To study the characteristics of APD

OBJECTIVE

To measure a number of important characteristics of an Avalanch photo Diode (APD).

1. Responsivity (R_o) of an APD.
2. Gain Versus Reverse bias voltage characteristics of an APD.

EQUIPMENT

- APD trainer kit
- Power Supply
- 9/125 μm SC To ST Fiber Optical Patch chords. (1 mtr)_02 nos.
- Digital / Analog ammeter having μA range
- DMM
- Optical Power Meter
- 10dB optical attenuator
- SC TO SC Adaptor

THEORY

1-1 Dark Current VS. Reverse Voltage

The APD dark current is categorized into surface leakage current I_{ds} flowing through the interface between the PN junction and InGa oxide layer and internal current I_{dg} generated inside the InGa substrate. (See Figure 1-2.)

Carriers not multiplied

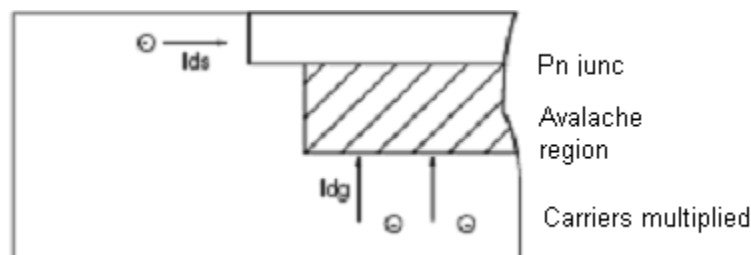


Figure 1-2 APD dark current

The surface leak current is not multiplied because it does not flow in the avalanche region, but the internally generated current flows in the avalanche region so it is multiplied. The total dark current I_D produced from an APD becomes as follows.

$$I_D = I_{Ds} + M \cdot I_{dg} \dots\dots\dots (1-4)$$

Where M is the multiplication ratio or gain Figure 1-3 graphically shows how the dark current changes with the reverse voltage. It is clear that I_{dg} increases as the reverse voltage rises.

1-2 Gain Vs. Reverse Voltage

The APD multiplication ratio (gain) depends on the electric field applied across the avalanche layer. Normally, the higher the reverse voltage, the higher the gain will be. However, if the reverse voltage is increased further, a voltage drop occurs due to the current flowing through the device series resistance and load resistance, causing the voltage applied to the avalanche layer to decrease. This means that the APD has a maximum gain whose value is dependent on the photocurrent. When the APD is operated near this maximum gain, the voltage drop tends to increase due to the series resistance component, resulting in an unwanted phenomenon in which the output photocurrent is not proportional to the amount of incident light. The APD gain also has temperature-dependent characteristics. The gain at a certain reverse voltage becomes small as the temperature rises. This is because the crystal lattice vibrates more heavily with an increasing temperature, and the accelerated carriers are apt to collide with the lattice before reaching an energy level sufficient to trigger ionization. To obtain a constant output, it is necessary to adjust the reverse voltage according to the changes in temperature or to keep the APD temperature constant. The temperature coefficient of gain is commonly expressed in $V/^{\circ}C$ or $\%/^{\circ}C$. When an APD is operated at a gain of 100, the temperature coefficient of the Reverse voltage will be almost equal to that of the breakdown voltage.

1-3 Spectral Response

Spectral response characteristics of APDs are almost the same as those of normal photodiodes if a reverse voltage is not applied. When a reverse voltage is applied, the spectral response curve will change slightly. This is because the multiplication efficiency of carriers injected into the avalanche region depends on the wavelength. This means that the gain changes depending on the incident light wavelength. It is therefore important to select an APD with spectral response Characteristics that match your application

1-4 Terminal Capacitances

APDs have the same terminal capacitance characteristics as those of normal photodiodes. To ensure high-speed response, it is necessary to apply a reverse voltage which makes the depletion layer thicker than the penetration depth of the light into the light absorption layer. If carriers are generated outside the depletion layer, they cause problems such as slow signal decay time. Since the terminal capacitance depends on the depletion layer thickness, it can be used as a guide to find to what extent the semiconductor substrate is depleted. Unlike the gain characteristics, the terminal capacitance is not temperature-dependent. However, it does vary depending on the reverse voltage, as shown in region (1) of Figure 1-4. So use the APD at a reverse voltage that ensures a constant terminal capacitance as in region (2)

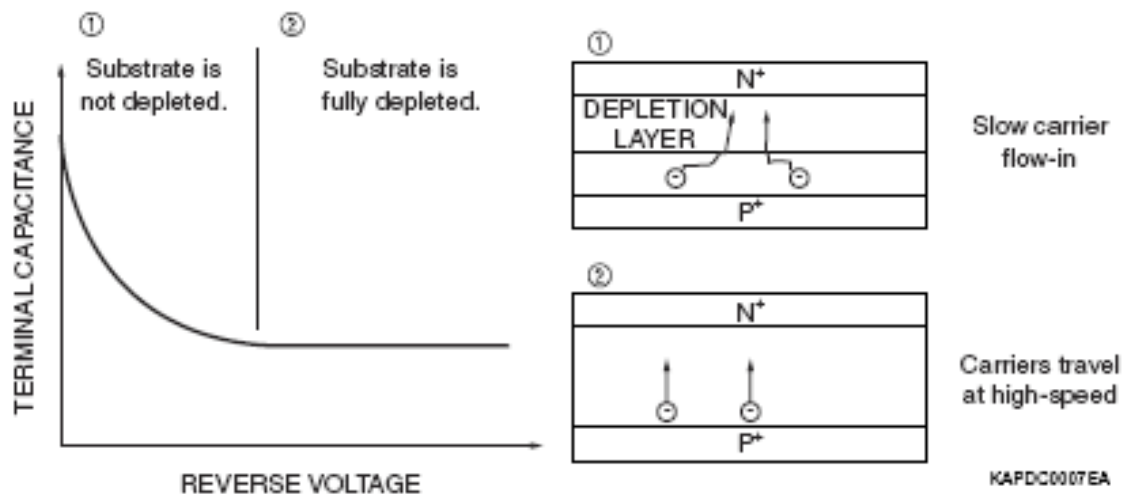


Figure 1-4 Terminal capacitances vs. reverse voltage

Background Information On PIN and APD Photodiode characteristics

Optical Detectors:

By definition photo detector converts light signals to electrical signals which can then be processed further. For fiber optic applications photo detectors work at standard wavelengths at 850, 1310 and 1550 nm. Suitable photodiodes may be either PIN diodes or avalanche photo diode (APD). In either case the operating wavelength determines the material used, for example Si being employed at 800-900nm and Ge or alloys of In, Ga, As and P at 1310nm.

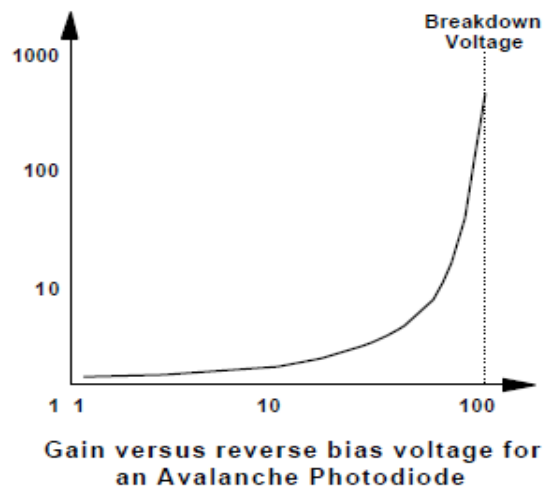
PIN diodes and APD's are variations on a basic depletion layer photodiode in which reverse current is altered by absorption of light at correct wavelength. APD's differ from PIN diode in that APD's have high gain so that with the correct circuitry better sensitivity can be achieved with APD's.

Definitions for photo diode:

Quantum Efficiency: It is defined for photo diodes as the fraction of incident photons having sufficient energy to liberate electrons. The symbol used is R and by definition it is dependent on both wavelength and photodiode material.

Responsivity: Responsivity of photo diode is a practical measure of output current for a given optical power input. It is defined as average output current divided by average incident optical power so its unit is A/W.

Gain: It is the multiplication of primary photon current. Gain exists in APDs only and is defined as ratio of the output current at an operating voltage to the current at low voltage where the gain is unity that is no multiplication occurs. In effect the primary photon current is multiplied by collision ionisation in a high field area. A typical gain characteristic is shown below.



Dark Current: In absence of light a small dark current flows in a photo diode which is caused by leakage in the reverse biased photodiode. It has a very small effect in the performance of a receiver in terms of sensitivity and is ignored in any further analysis. It has been shown that dark current less than 0.1nA have very little effect. Finally photodiodes normally exhibits rise and fall times less than 1nSec. In APD's depletion capacitance increases rapidly below a certain reverse bias voltage, so to maintain fast rise and fall times a minimum bias voltage is maintained.

PROCEDURE

General settings for characteristic measurements

1. To perform this experiment a stable reference source of light at about 1310nm is required.
2. The 1310 nm laser current can be set by a potentiometer on the front panel of the unit. A pair of DPM is also provided to measure the current through and voltage across the laser.
3. Connect power supply to the Laser source and switch it ON.
4. Connect Laser to the optical power meter using the fiber patch chord provided and adjust the optical power to 10 μ W. while adjusting this make sure that laser must be in lasing mode.(if you are using FOM-DUAL 1310 laser source then use 10dB attenuator to reduce the power to 10 μ W.)
5. Note down the exact optical power for reference. Ensure that fiber optic power meter is set for operation of 1310nm, by pressing the λ key.
6. **The optical power level set is now the reference power for rest of the exercise and should not be altered again. Make sure also that the fiber is not under stress or tightly wound when taking the reference reading as this could make the reference setting invalid.**

a. Responsivity (R_o) of an APD.

7. Connect power supply with proper polarity to APD and switch it ON.
8. Connect laser source which is set at 10 μ W to APD using patch chord provided.(use attenuators if required)
9. Set APD biasing voltage to 10V using pot P1 and P2. Measure the bias voltage at Bias voltage test point with respect to ground.

**Note: To adjust low APD bias voltages rotate pot P1 fully clockwise position and then adjust Pot P2 to get the desired voltage
To adjust high APD bias voltages rotate pot P1 fully anticlockwise position and then adjust Pot P2 to get the desired voltage**

10. For an APD the Responsivity characteristics are carried out at low voltages to ensure that no gain is present.
11. Short the jumper J1 to 1-2 position

12. APD is contained in metal case and in series with 100K resistor.
13. The current in the APD can be measured by connecting an ammeter in series with the load resistor R_L using the connecting point provided.
14. Note down the ammeter reading (which is generally in μA range). Verify the Responsivity of the APD with datasheet.

b. Gain Versus Reverse bias voltage characteristics of an APD.

15. Keep the same experiment setting as per Responsivity measurement, and note down the APD current and Bias voltage. Recording this value as "NO GAIN " current
16. Now go on increasing the APD bias voltage in step of 2V and note down the APD current.
17. Note that at some stage the output current will increase rapidly for small change in applied bias voltage. This means that the bias voltage is close to the breakdown voltage for the device. In our case do not exceed the 54V to 55V. **Do not exceed this operating point as damage to the APD could result.**
18. When the measurements are completed, find the gain at each bias voltage by dividing the APD output current at that bias voltage by the NO GAIN value recorded above. Hence plot the gain versus voltage characteristics for the APD on graph paper.

OBSERVATIONS:

Note: The observations given below are the sample readings. Readings may differ from kit to kit.

APD Characteristics

a. Responsivity (R_o) of an APD.

Test conditions:

1. 1310 Laser output power = 10 μW . use 10dB attenuator.
2. APD bias voltage = 10V.
3. Measured current through load resistor = 7.8 μA

Responsivity (_{practical}) = 0.78 $\mu A / \mu W$. in other words it is 0.78 A / W.

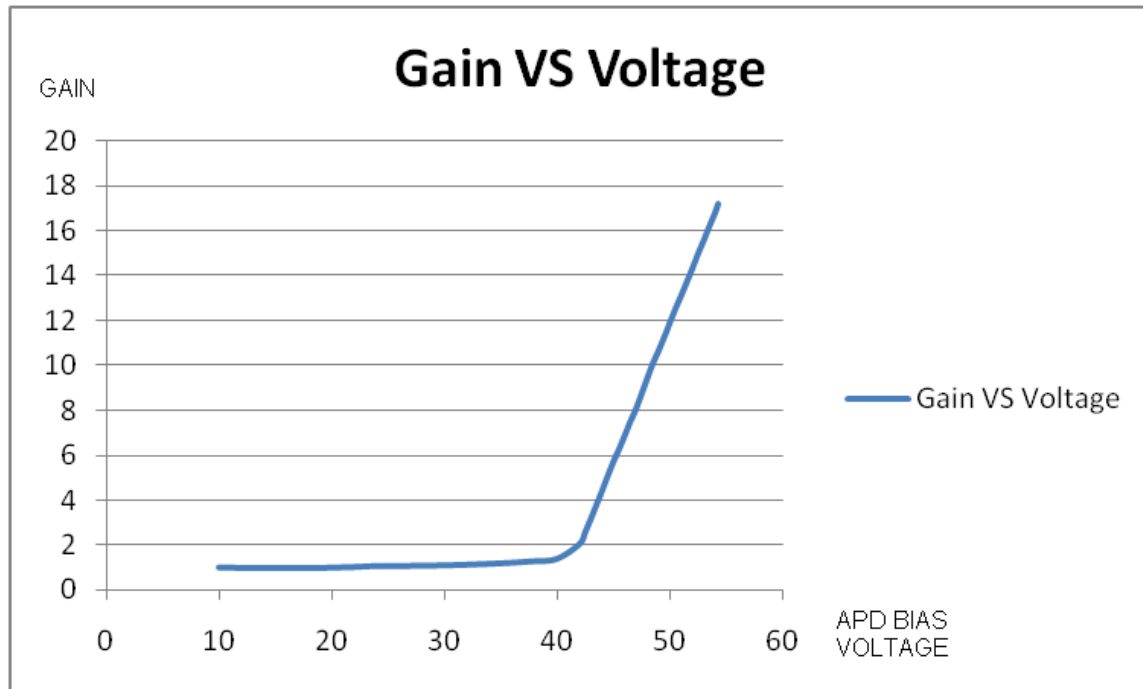
Responsivity (_{given}) = 0.75 $\mu A / \mu W$. in other words it is 0.75 A / W.

b. Gain Versus Reverse bias voltage characteristics of an APD.

Test conditions:

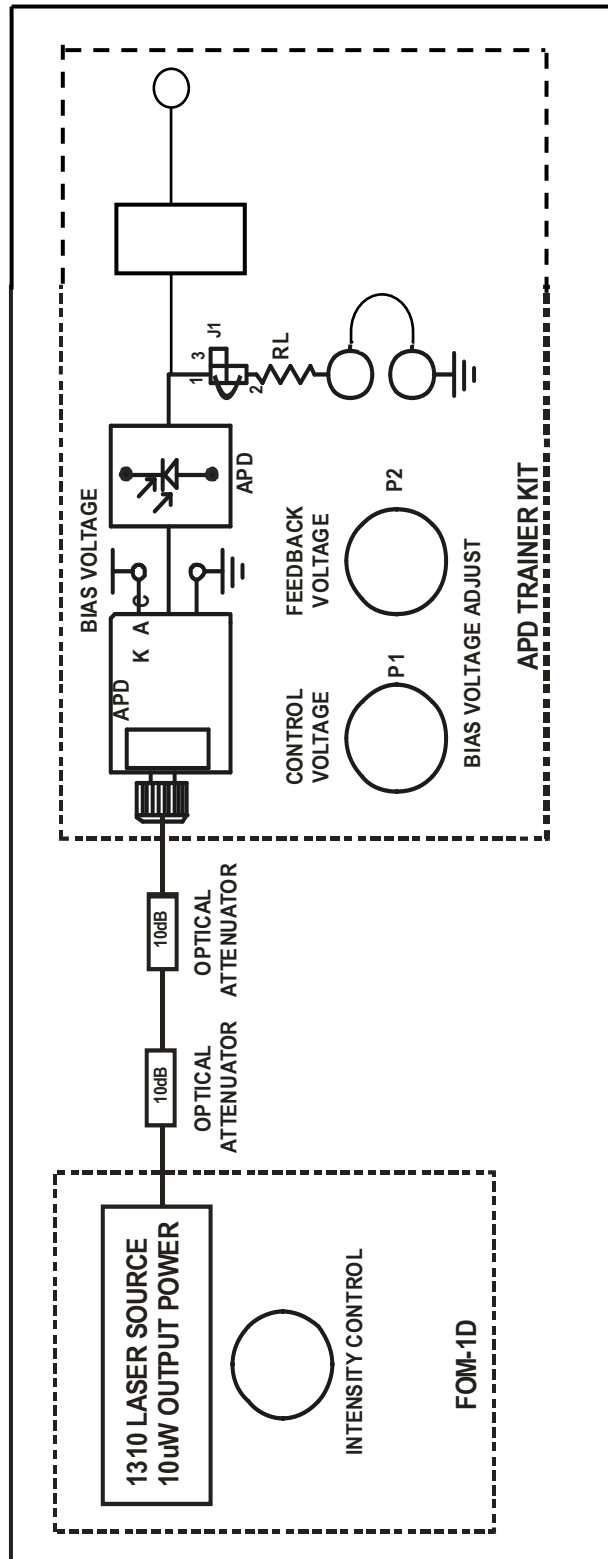
1. 1310 Laser output power = 10 μW . use 10dB attenuator.

SR NO	APD Biasing Voltage. (V)	APD current (μ A)	Voltage VS current gain
1.	10	7.8	1
2.	12	7.70	0.98
3.	14	7.70	0.98
4.	16	7.58	0.97
5.	18	7.60	0.97
6.	20	7.79	0.99
7.	22	8.01	1.02
8.	24	8.37	1.07
9.	26	8.28	1.06
10.	28	8.50	1.08
11.	30	8.57	1.09
12.	32	8.75	1.12
13.	34	9.07	1.16
14.	36	9.44	1.21
15.	38	10.01	1.28
16.	40	10.82	1.38
17.	42	15.92	2.04
18.	42.5	20.07	2.57
19.	43	24.67	3.16
20.	43.5	29.58	3.79
21.	44	34.40	4.41
22.	44.5	39.65	5.08
23.	45	44.50	5.70
24.	45.5	48.75	6.25
25.	46	53.52	6.86
26.	46.5	58.38	7.48
27.	47	62.75	8.04
28.	47.5	67.99	8.71
29.	48	72.44	9.40
30.	48.5	78.55	10.07
31.	49	82.93	10.63
32.	49.5	87.61	11.23
33.	50	92.61	11.87
34.	50.5	97.51	12.50
35.	51	102.07	13.08
36.	51.5	106.73	13.68
37.	52	111.50	14.29
38.	52.5	116.55	14.94
39.	53	121.06	15.52
40.	53.5	126.11	16.16
41.	54	130.66	16.75
42.	54.3	133.97	17.17

**CONCLUSION:**

Note down all the readings as mentioned in above procedure. Compare the Responsivity values with the APD datasheet provided. Also verify the biasing voltage versus gain characteristics.

EXPERIMENT NO.2



EXPERIMENT NO.2

NAME

To set up TTL link using APD

OBJECTIVE

To received TTL signal (APD).

EQUIPMENT

- APD trainer kit
- Power Supply
- 9/125 μm SC TO ST Fiber Optical Patch chord. (1 mtr)- 02 nos.
- Digital / Analog ammeter having μA range
- DMM
- Optical Power Meter
- 10dB optical attenuator
- SC TO SC adaptor
- 1310 Lezer source with TTL transmitter

PROCEDURE

Set up TTL link using APD

1. To perform this experiment 1310nm source with TTL transmitter required.
2. Connect power supply to the Laser source and switch it ON.
3. Connect Laser to the optical power meter using the fiber patch chord provided and adjust the optical power to $10\mu\text{W}$.
4. Note down the exact optical power for reference. Ensure that fiber optic power meter is set for operation of 1310nm, by pressing the λ key.
5. Apply the TTL input to the 1310nm lezer source.
6. Connect power supply with proper polarity to APD and switch it ON.
7. Set the at minimum biasing voltage.
8. Connect laser source which is transmit TTL input to APD using patch chord provided.(use10dB attenuators)
9. Set the jumper J1 at 1-2 position.
10. Connect link between pcs2 as shown in fig.
11. Check TTL o/p at output test point.

OBSERVATIONS:

Note: The observe TTL output.

